

2025

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : DSCC-8P

(Object Oriented Programming using Java)

Full Marks : 25

Set - 1

Marks Distribution

Source Code :	10
Output :	05
Laboratory Notebook :	05
Viva voce :	05
	25

Answer *any one* question.

1. Write a Java program to accept a string and change the case of each letter of the string. Display the new string.

Sample Input : Hello sTRing Tutorial

Sample Output : hELLO StrING tUTORIAL.

2. Create an abstract class Vehicle with properties : vehicleNumber, vehicleName, and fuelType. It should include an abstract method calculateRange() and a concrete method display() to show basic details. Create two subclasses : Car and Motorcycle, each with fuelCapacity and mileage. Implement the calculateRange() method using the formula :

$$\text{Range} = \text{fuelCapacity} \times \text{mileage}$$

Override the display() method to include all details and the calculated range. Write a test class to demonstrate the functionality of both classes.

3. Write a Java program to print all permutation of a given string of length 3. Permutations of the string 'ABC' is — ABC, ACB, BAC, BCA, CBA, CAB.
4. Write a Java program to calculate area of Triangle using class TriangleArea with the following methods :
- (a) Initialize the base and height of a triangle using constructor
 - (b) Give method to calculate area() using base and height
 - (c) Give method to print the triangle details, i.e., its base, height and area.
5. Write a Java program that reads a list of integers from the user and throws an exception if any numbers are duplicates.
6. Write a Java program to create a class called 'Book' with private instance variables title, author, and price. Provide public getter and setter methods to access and modify these variables. Add a method called applyDiscount() that takes a percentage as a parameter and reduces the price by that percentage.
7. Develop a Java program to create an interface Resizable with methods resizeWidth(int width) and resizeHeight(int height) that allow an object to be resized. Create a class Rectangle that implements the Resizable interface and implements the methods.
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Full Marks : 25

Set - 2

Marks Distribution

Source Code :	10
Output :	05
Laboratory Notebook :	05
Viva voce :	05
	25

Answer *any one* question.

- Write a Java program to implement a Queue using a custom class **MyQueue**. The class should have data members : front, rear, and an array to store the queue elements. Implement the following member functions :
 - **enqueue()** — to insert an element into the queue
 - **dequeue()** — to remove an element from the queue
 - **display()** — to show all elements of the queue along with its current status
 Also, define a default constructor to initialize the queue's data members.

- Design a Java program to simulate a vehicle engine system. The engine should have two attributes : speed (of type int) and gear (of type int). Define an **interface** containing the following methods :
 - **speedUp(int value)** — to increase the engine speed
 - **changeGear(int value)** — to change the gear of the vehicle. Also ensure that gear value must be between 1 to 6.
 Create a class that **implements the interface** and provides proper definitions for both methods.

3. Write a Java program that creates a class hierarchy for employees of a company. The base class should be Employee, with subclasses Manager, Developer, and Programmer. Each subclass should have properties such as name, address, salary, no_of_projects and job title. Implement methods for calculating bonuses, generating performance reports, and managing projects.

 4. Write a Java program to create an abstract class Shape with abstract methods calculateArea() and calculatePerimeter(). Create subclasses Circle and Triangle that extend the Shape class and implement the respective methods to calculate the area and perimeter of each shape.

 5. Write a Java program using threads to print numbers from 1 to 5 in two separate threads. One thread should print the odd numbers, and the other thread should print the even numbers. Thus, one thread prints the odd numbers [1, 3, 5]. Another thread prints the even numbers [2, 4].

 6. Write a Java program to create your own exception class called as NegativeNumberException whenever negative number is entered by user.

 7. Write a Java program that checks if a given string is palindrome. A palindrome is a string that reads the same forward and backward. Now take a string input from the user and check if it is a palindrome. Ignore space and punctuation.
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COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : DSCC-8P

(Object Oriented Programming using Java)

Full Marks : 25

Set - 3

Marks Distribution

Source Code :	10
Output :	05
Laboratory Notebook :	05
Viva voce :	05
	25

Answer *any one* question.

1. Write a Java program to create and fill shapes using applet.
2. Write a Java program that defines a custom exception named NegativeException. The program should throw this exception whenever a negative value is entered into an array. Implement appropriate exception handling to display an error message when such a value is detected.

3. Write a Java program that accepts a person's full name as input and converts it into an initial format, displaying only the initials of the first names followed by the full surname.

Example :

Input : Arjun Sachin Ramesh Tendulkar

Output : A.S.R.Tendulkar.

4. Write a Java program that reads a text file containing a series of numbers. The program should extract all even numbers from the file and write them into a separate output file. Finally, display the contents of the output file on the screen.

5. Write a Java program to create class Product (pid, price, quantity) with parameterized constructor.

Create a main function in different class (say XYZ) and perform following task :

- (a) Accept five product information from user and store in an array.
- (b) Find Pid of product with highest price.
- (c) Create method (with array of product's object as argument) in XYZ class to calculate and return total amount spent on all products. (amount spent on single product=price of product * quantity of product).

6. Write a Java program to create a base class called "vehicle" that stores number of wheels and speed.

Create the following derived classes :

"car" that inherits "vehicle" and also stores number of passengers.

"truck" that inherits "vehicle" and also stores the load limit. Write a main function to create objects of these two derived classes and display all the information about "car" and "truck". Also compare the speed of these two vehicles — car and truck and display which one is faster.

7. Write a Java program to create an abstract class Accounts with the following details :

Data Members :

(a) Balance (b) accountNumber (c) accountHoldersName (d) address

Methods :

(a) Abstract method : withdraw()

(b) Abstract method : deposit()

(c) display() to show the balance of the account number

Create a subclass of this class SavingsAccount and add the following details :

Data Members :

(a) rateOfInterest

Methods :

(a) calculateAmount()

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Set - 4

Marks Distribution

Source Code :	10
Output :	05
Laboratory Notebook :	05
Viva voce :	05
	25

Answer *any one* question.

1. Write a Java program that implements a Stack class for integers. Write the basic operations of Stack like push(), pop(), overflow(), underflow().
2. Write a Java program to accept a decimal number from keyboard and convert it into Binary and Octal base system. Read the decimal number as a private member of class Conversion.
3. Write a Java program to create abstract class Shape. Implement two subclasses called Circle, Rectangle and print their respective area by overriding base class methods.

4. Write a Java program that create a class Bank with the following menu :

- (a) Display all account details
- (b) Search by account number
- (c) Deposit the amount
- (d) Withdraw the amount
- (e) Exit.

5. Write a Java program to create a method that takes a string as input and throws an exception if the string contains vowels.

6. Write a Java program to create a class Voter(voterId, name, age) with parameterized constructor. The parameterized constructor should throw a checked exception if age is less than 18. The message of exception is "invalid age for voter".

7. Write a Java program to display the clock using applet.
